

Diffusion:-

- the net movement of particles from a region of higher concentration to a region of lower concentration, as a result of their random movement.
- the energy for diffusion comes from the kinetic energy of random movement of molecules and ions.

Diffusion in cells:-

- some substances move in and out of the cell by diffusion through the cell membrane.

Why is it important for living things?

- it allows them to gain the useful substances for energy and growth and get rid of waste products.

Factors that influence diffusion:-

- when a cell's surface area increases, the amount of diffusion increases.
- higher the temperature, higher the movement of molecules, higher the rate of diffusion.
- steeper the concentration gradient, faster the rate of diffusion.
- bigger the distance, slower the rate of the diffusion.

Osmosis:

- net movement of particles from a region of higher water potential to a region of lower water potential, across a partially permeable membrane.

how does water enter the cell?

- by osmosis through the cell membrane.

Investigate the effects on plant tissues of immersing them in solutions of different concentrations:

- * when you immerse plant tissues in a solution of lower water potential, the water in those cells leaves by osmosis to its surrounding.

└ The cytoplasm pulls away from the cell wall
└ the cell becomes flaccid (plasmolysis)

- * when you immerse plant tissues in a solution of higher water potential, the water in the solution enters the cells by osmosis.

└ The cytoplasm pushes hard against the cell wall
└ the cell becomes turgid (turgor pressure)

Osmosis and dialysis tubing:

- dialysis tubing is a semipermeable membrane
- * if we have a solution of salt and water, then the salt ions cannot pass through the membrane but water can. This is a sign of osmosis.

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How are plants supported:-

- by the pressure of water inside the cells pressing outwards on the cell wall

the importance of water potential/osmosis in the uptake and loss of water by organisms:-

- (in plants) the roots are surrounded by soil water and the cytoplasm of root cells has a lower water potential than the soil water.
- * The soil water is absorbed by the root hairs.

role of water as a solvent:-

- * transport of nutrients (e.g. glucose / amino acids in blood)
- * removal of excretory products (e.g. urea / ammonia)
- * process of digestion (e.g. food dissolved in water)

ACTIVE TRANSPORT

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Active transport:-

- the movement of particles through a membrane from a region of low concentration to a region of high concentration, using energy from respiration.

active transport in root hairs:-

- root hair cells take in nitrate ions
- nitrate ions in plant root hair cells have a higher concentration than soil.
- diffusion gradient would take the nitrate ions from the roots
- still root hair cells take in nitrate ions through a process called 'active transport'
- protein carriers move molecules or ions across membranes, including ion uptake by root hairs.